Mathematical Models in Biotechnology

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Abstract

Intracellular fluxes, kinetic parameters, and metabolite concentrations are sparsely measured at genome scale. Flux balance analysis estimates fluxes from network stoichiometry, reaction bounds, and an optimization objective. This chapter derives the mass-balance constraints from mole balances and states the assumptions that lead to their steady-state form. It then combines these equations with reaction bounds based on reversibility, enzyme capacity, substrate saturation, enzyme abundance, and regulation. A urea-cycle model shows how public thermodynamic and kinetic estimates set reaction bounds. A feedback-inhibited pathway shows how expression and allosteric regulation change a capacity bound. Two larger studies illustrate related methods in cell-free protein synthesis and strain design. Together, these examples show how condition-specific data and proposed interventions change flux predictions through reaction bounds.

1 Introduction

Biotechnology depends on predicting and directing cellular metabolism. Metabolic engineering redirects flux toward a product. Bioprocess development selects operating conditions that sustain production. Strain design identifies genetic changes that alter the phenotype. Each task requires an estimate of metabolic reaction rates under a specified condition [1], although these rates are difficult to measure directly. Genome sequences and reaction annotations provide a network structure, but reconstructions remain uncertain in reaction membership, directionality, compartments, cofactors, and biomass composition. Fluxes, kinetic parameters, and metabolite concentrations are even less complete at genome scale, so models must estimate phenotype from incomplete measurements. Bioprocess models address this problem at several levels of biological detail. Unstructured models track biomass, extracellular substrates, and products with lumped growth, uptake, and production rates. Monod growth, inhibition, maintenance, and growth-associated production laws support batch, fed-batch, and continuous reactor analysis, but do not resolve intracellular fluxes [2]. Structured kinetic models add intracellular pools, enzymes, or cellular machinery. Early single-cell models from Shuler and coworkers coupled intracellular metabolism to extracellular nutrients in microbial and mammalian cells [3, 4]. Reaction-level kinetic models assign mass-action, Michaelis–Menten, or power-law rates to individual reactions and integrate the resulting differential equations [5–8]. These models describe transient intracellular behavior. As their resolution increases, however, so does the number of required kinetic parameters. This parameter burden motivates formulations that do not specify a kinetic law for every reaction.

Unstructured and structured kinetic models calculate rates from specified empirical or mechanistic relationships. Constraint-based models instead define the flux distributions allowed by network stoichiometry, reaction bounds, and other physical or biological constraints. Flux balance analysis

(FBA) uses an objective to select a solution from this feasible set [9, 10]. Because a constraint-based model does not require a kinetic rate law for every reaction, it can represent much larger networks, including genome-scale reconstructions [1, 11]. Hybrid cybernetic models provide a dynamic bridge between the kinetic and constraint-based descriptions by combining metabolic modes with cybernetic variables that allocate enzyme synthesis and activity among those modes. Kim, Varner, and Ramkrishna applied this framework to anaerobic *E. coli* using elementary flux modes [12]. Vilkhovoy, Minot, and Varner later developed hybrid cybernetic modeling with flux balance analysis (HCM-FBA). This method uses FBA solutions in place of exhaustive elementary-mode enumeration and therefore extends the approach to larger networks [13]. Martínez and coworkers subsequently used a hybrid cybernetic model with a reduced set of elementary modes to describe dynamic CHO-S metabolism [14].

Rather than specifying every relationship mechanistically, data-driven models learn patterns directly from biological time-series data. Long short-term memory (LSTM) and gated recurrent unit (GRU) networks are established recurrent architectures for learning temporal relationships [15, 16]. The relatively recent structured state-space sequence models S4 and S5 provide efficient representations of longer time dependencies [17, 18]. Neural ordinary differential equations learn continuous-time state dynamics from data [19]. These approaches are being explored across fermentation, cell culture, and other bioprocesses, often as forecasting models, soft sensors, or data-driven components of hybrid mechanistic models [20]. Chinese hamster ovary (CHO) cell culture is an important application because of its central role in monoclonal antibody production. Recent CHO studies have used machine learning to optimize cultivation conditions [21], optimize media and predict critical quality attributes [22], and predict antibody glycosylation from daily spent-medium measurements [23]. Recent reviews place these efforts within broader mechanistic and hybrid modeling strategies [24]. Data-driven models can learn relationships that are difficult to specify mechanistically, but they do not inherently conserve mass or enforce reaction feasibility. Hybrid models can address this limitation by combining learned relationships with the mechanistic balances and biological constraints.

Within this broader modeling landscape, this chapter focuses on how thermodynamic, kinetic, expression, and regulatory data can be incorporated into constraint-based models through reaction bounds. We first derive the mass-balance constraints from mole balances and identify the assumptions required for their steady-state form. We then formulate the linear program and describe how reversibility, enzyme capacity, substrate saturation, enzyme abundance, and biological regulation restrict the feasible flux space. Two worked examples illustrate these ideas in a urea-cycle network and a feedback-inhibited pathway. Larger studies in dynamic cell-free protein synthesis [25, 26] and model-guided strain design [27, 28] show how the same principles extend to practical systems. Together, these examples show how condition-specific measurements and proposed interventions change predicted metabolic fluxes.

2 From Mole Balances to Flux Constraints

The mass-balance constraints used in FBA follow from mole balances, and their derivation defines both the normalization basis and the steady-state assumptions. Consider a well-mixed culture with biomass concentration B and reactor volume $\bar{V}$. The biomass basis is therefore given by:

$$\mathcal{B} \equiv B\bar{V}, \tag{1}$$

where B has units gDW/L, $\bar{V}$ has units L, and $\mathcal{B}$ is the total biomass in gDW. An intracellular concentration C_i has units mmol/gDW, and an intracellular flux $\hat{v}_j$ has units mmol/gDW/h. Multiplication by the biomass basis $\mathcal{B}$ gives the total intracellular amount $C_i\mathcal{B}$ and total reaction rate

$\hat{v}_j\mathcal{B}$. A general intracellular balance includes reaction, active transport, and physical transport. Let $c_{i,s}$ be the concentration of species i in a hypothetical bulk stream s, and let $\dot{V}_s$ be its volumetric flow rate. The intracellular mole balance is given by:

$$\sum_s d_s c_{i,s} \dot{V}_s + \mathcal{B} \sum_j \sigma_{ij} \hat{v}_j = \frac{d}{dt}\big(C_i \mathcal{B}\big), \tag{2}$$

where the stream-direction coefficient $d_s = +1$ for an inlet and $d_s = -1$ for an outlet. The coefficient σ_{ij} is entry (i, j) of the stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{m \times n}$ for m metabolites and n reactions. Each term has units mmol/h. The stream concentration $c_{i,s}$ has units mmol/L, whereas C_i has units mmol/gDW. Because the cell membrane excludes bulk physical flow from the intracellular control volume, the volumetric flow rate $\dot{V}_s = 0$ and the first term vanishes. Active transport remains in the flux vector $\hat{\mathbf{v}}$ because it moves molecules without bulk volume flow.

Expanding the accumulation term before assuming a fixed reactor volume and substituting the biomass-basis relation $\mathcal{B} = B\bar{V}$ gives:

$$\frac{d}{dt}\big(C_i \mathcal{B}\big) = B\bar{V}\frac{dC_i}{dt} + C_i\bar{V}\frac{dB}{dt} + C_i B\frac{d\bar{V}}{dt}. \tag{3}$$

After division by the biomass basis $\mathcal{B} = B\bar{V}$, Equation (2) becomes:

$$\frac{1}{\mathcal{B}}\sum_s d_s c_{i,s}\dot{V}_s + \sum_j \sigma_{ij}\hat{v}_j = \frac{dC_i}{dt} + \frac{C_i}{B}\frac{dB}{dt} + \frac{C_i}{\bar{V}}\frac{d\bar{V}}{dt}. \tag{4}$$

For a batch or fed-batch culture with no biomass flow, total biomass changes through growth, so the specific growth rate is given by:

$$\mu \equiv \frac{1}{\mathcal{B}}\frac{d\mathcal{B}}{dt} = \frac{1}{B}\frac{dB}{dt} + \frac{1}{\bar{V}}\frac{d\bar{V}}{dt}. \tag{5}$$

The membrane removes the physical-flow term from Equation (4). With the concentration vector defined as $\mathbf{x} = (C_1, \ldots, C_m)^\top$ in mmol/gDW, combining the remaining terms with Equation (5) gives the dynamic intracellular balance:

$$\mathbf{S}\hat{\mathbf{v}} = \frac{d\mathbf{x}}{dt} + \mu\mathbf{x}. \tag{6}$$

Balanced growth gives $d\mathbf{x}/dt = \mathbf{0}$, so this balance reduces to:

$$C_i\mu = \sum_j \sigma_{ij}\hat{v}_j \qquad \Longleftrightarrow \qquad \boxed{\mathbf{S}\hat{\mathbf{v}} = \mu\mathbf{x}}, \tag{7}$$

where the flux vector $\hat{\mathbf{v}} \in \mathbb{R}^n$ contains the reaction fluxes in mmol/gDW/h. Net production therefore balances growth dilution. For a fixed-volume batch reactor, $d\bar{V}/dt = 0$ and $\mu = (1/B)dB/dt$. For fed-batch operation, $d\bar{V}/dt \neq 0$, so the volume term must remain. The fixed-volume result is a special case of the general balance.

Conventional FBA neglects the dilution term and uses the following approximation:

$$\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}. \tag{8}$$

This approximation is appropriate when the growth-dilution term $\mu\mathbf{x}$ is small relative to the metabolic fluxes [9]. Setting the growth rate $\mu = 0$ is not sufficient. A cell-free batch has no growing biomass, but its transcript, protein, and metabolite concentrations may still change. Its accumulation term $d\mathbf{x}/dt$ must remain. On a fixed concentration basis, Equation (6) becomes $\mathbf{S}\mathbf{v} = d\mathbf{x}/dt$. Here $\mathbf{v}$ is the cell-free flux vector. Dynamic constraint-based models of cell-free protein synthesis have used this form to impose time-dependent concentration changes over short intervals [25]. The same balance underlies each model class, but each closes the reaction term differently. Unstructured models use lumped rates, structured kinetic models use state-dependent intracellular rates, and constraint-based models use bounds and an objective. Constraint-based reconstructions represent uptake and secretion with transport and exchange reactions. A boundary reaction $\emptyset \rightleftharpoons M_i$ enters the stoichiometric matrix $\mathbf{S}$ as an isolated nonzero column with no intracellular reaction partner [1]. Its flux describes molecular transport across the modeled boundary rather than a reactor stream. The included boundary reactions and their bounds define the modeled system boundary.

3 The Linear Program and Its Geometry

The balance equations conserve mass but usually admit many flux distributions. Flux bounds restrict this set to a feasible polytope. An objective then selects an optimal vertex or face (Fig. 1). FBA adds lower and upper bounds, represented by the lower-bound vector $\boldsymbol{\ell}$ and upper-bound vector $\mathbf{u}$, and a linear objective with coefficient vector $\mathbf{c}$. Together, the mass-balance equations and the bounds $\boldsymbol{\ell} \leq \hat{\mathbf{v}} \leq \mathbf{u}$ form the flux constraint system. The objective $\mathbf{c}^\top \hat{\mathbf{v}}$ then selects a solution from that system. Because bounds can represent reaction direction, enzyme capacity, and substrate availability, the balances, bounds, and quantity to maximize together define the linear program:

$$\begin{aligned} \max_{\hat{\mathbf{v}} \in \mathbb{R}^n} \quad & \mathbf{c}^\top \hat{\mathbf{v}} \\ \text{subject to} \quad & \mathbf{S}\hat{\mathbf{v}} = \mathbf{0} \quad (\text{or } \mu\mathbf{x}), \\ & \boldsymbol{\ell} \leq \hat{\mathbf{v}} \leq \mathbf{u}. \end{aligned} \tag{9}$$

This standard FBA formulation [9] separates the constraint system from the modeling choice expressed by the objective. Setting the objective coefficient $c_j = 1$ for a biomass pseudo-reaction maximizes growth by consuming precursors in the proportions needed to form new cell mass. Setting $c_j = 1$ for a product-export reaction instead maximizes product secretion. The constraints remain unchanged. Only the objective coefficient vector $\mathbf{c}$ differs. The growth-aware form remains linear when the concentration vector $\mathbf{x}$ is measured and the growth rate μ is fixed or enters as a scalar decision variable. If both the growth rate μ and concentration vector $\mathbf{x}$ are unknown, their product is bilinear and the problem is no longer an LP. A model that retains the growth-dilution term $\mu\mathbf{x}$ must also define its relation to the biomass pseudo-reaction so that growth-associated demands are not counted twice. The conservation equations usually admit many flux vectors. The corresponding null space is given by:

$$\mathcal{N}(\mathbf{S}) = \{\hat{\mathbf{v}} \in \mathbb{R}^n : \mathbf{S}\hat{\mathbf{v}} = \mathbf{0}\}.$$

This set contains every steady-state flux vector and contains more than one vector when the rank condition $\text{rank}(\mathbf{S}) < n$ holds. Bounds restrict the null space to a feasible polytope. The objective selects a vertex or face. A face can contain multiple optimal flux vectors, so optimization does not always give a unique solution.

The singular value decomposition identifies both null spaces by factoring the stoichiometric matrix as $\mathbf{S} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$, where $\mathbf{U}$ is the left-singular-vector matrix. The matrix $\boldsymbol{\Sigma}$ contains the singular values on its diagonal. The matrix $\mathbf{V}$ is the right-singular-vector matrix. Columns of the

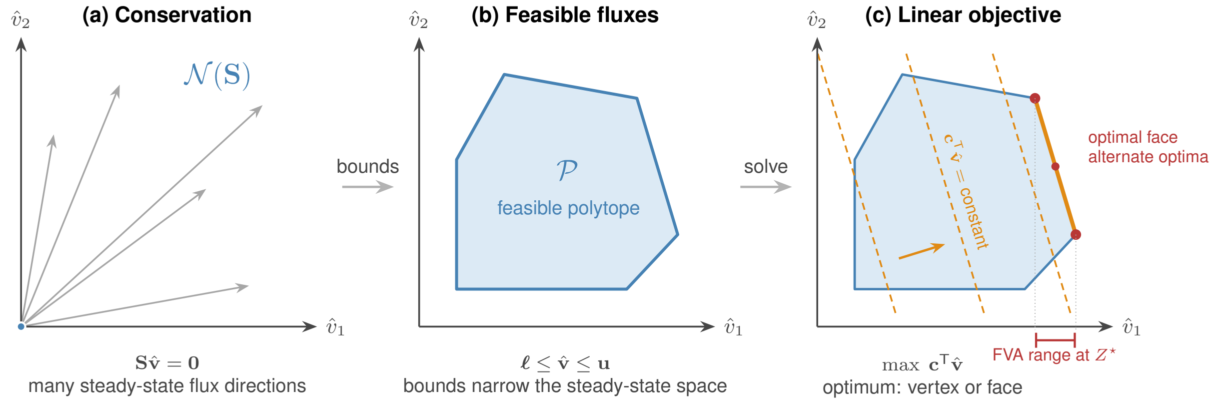

Figure 1: Geometry of FBA in a two-flux projection. (a) Conservation restricts steady-state fluxes to the null space $\mathcal{N}(\mathbf{S})$. (b) Lower and upper bounds form the feasible polytope $\mathcal{P}$. (c) The objective moves toward larger values until it reaches an optimal vertex or face. Fluxes can vary along an optimal face without changing the objective. FVA reports the resulting interval for each reaction, shown here for reaction flux $\hat{v}_1$.

right-singular-vector matrix $\mathbf{V}$ associated with zero singular values span the right null space. Every steady-state flux vector is a linear combination of these columns. Columns of the left-singular-vector matrix $\mathbf{U}$ associated with zero singular values span the left null space. A coefficient vector $\mathbf{y}$ in this space defines the conserved quantity $\mathbf{y}^\top\mathbf{x}$, such as the total concentration of a cofactor pair. Genome-scale models often have a large right null space because the rank condition $\text{rank}(\mathbf{S}) < n$ holds [1]. Flux variability analysis (FVA) measures the remaining freedom in each reaction [10]. FVA fixes the objective at its optimum, or at a chosen fraction of that optimum, and then minimizes and maximizes each flux. Equal minimum and maximum values mean that a reaction is fixed on that optimal face. A wide interval means that the reaction can vary without changing the required objective value. A different objective can select a different face and produce different intervals.

As condition-specific bounds are added, the feasible set may become empty. In that case, conflicting bounds or mass-balance constraints should be traced back to their data sources. Temporary slack variables can help diagnose the conflict, but a relaxed solution should not be mistaken for a biologically feasible prediction.

4 Integrative Flux Bounds

Reaction bounds can include several types of condition-specific data. For reaction j, the bound is written as:

$$-\delta_j\left[V^\circ_{\max,j}\,(e/e^\circ)\,\theta_j(\cdot)\,f_j(\cdot)\right]\ \leq\ \hat{v}_j\ \leq\ V^\circ_{\max,j}\,(e/e^\circ)\,\theta_j(\cdot)\,f_j(\cdot), \tag{10}$$

where $V^\circ_{\max,j}$ is a reference capacity with units of flux and the remaining factors are dimensionless. The reversibility factor δ_j sets the reverse direction. The reference capacity $V^\circ_{\max,j}$ and saturation factor f_j describe enzyme capacity and substrate saturation. The enzyme-abundance ratio (e/e°) compares the current enzyme concentration e with the reference concentration e°. The allosteric factor θ_j describes enzyme activity. Transcriptional and translational controls u_j and w_j act through the expression model and change the enzyme-abundance ratio (e/e°). They are distinct from the allosteric factor θ_j. When data are unavailable, the corresponding factor is set to one.

Measurements or mechanistic models can then replace that default. Equation (10) uses the same reference magnitude in both directions as a simplifying baseline. A reversible enzyme can instead have distinct forward and reverse capacities.

The reversibility factor δ_j has two values:

$$\delta_j \;=\; \begin{cases} 1, & \text{reaction } j \text{ reversible}, \\ 0, & \text{reaction } j \text{ irreversible}, \end{cases} . \tag{11}$$

When the reversibility factor $\delta_j = 1$, the lower bound is negative and reverse flux is allowed. When the reversibility factor $\delta_j = 0$, the lower bound is zero. A simple assignment compares the standard transformed reaction Gibbs energy $\Delta_\mathrm{r} G_j'^\circ$ with a heuristic reversibility threshold $\Delta_\mathrm{r} G_j'^*$. A full thermodynamic analysis instead propagates stated activity ranges and uncertainties through the reaction Gibbs energy, which is given by:

$$\Delta_\mathrm{r} G_j' = \Delta_\mathrm{r} G_j'^\circ + R T_\mathrm{abs} \ln Q_j, \qquad Q_j = \prod_i a_i^{\sigma_{ij}}, \tag{12}$$

where the gas constant $R = 8.314 \times 10^{-3}\ \mathrm{kJ\,mol^{-1}\,K^{-1}}$. The quantity T_abs is the absolute temperature in K. The quantity Q_j is the dimensionless reaction quotient. The quantity a_i is the dimensionless activity of species i relative to its standard state. The coefficient σ_{ij} is the stoichiometric coefficient of species i in reaction j. It is negative for reactants and positive for products. The prime denotes a biochemical transformed quantity evaluated under specified reference conditions, including pH and ionic strength. Reaction direction is tested by whether the reaction Gibbs energy $\Delta_\mathrm{r} G_j'$ can change sign over the allowed activity ranges. Forward flux requires $\Delta_\mathrm{r} G_j' < 0$. Reverse flux requires $\Delta_\mathrm{r} G_j' > 0$. At equilibrium, $\Delta_\mathrm{r} G_j' = 0$ [29]. The component contribution method provides consistent standard transformed Gibbs-energy estimates and uncertainties [30]. eQuilibrator makes these estimates available for metabolic calculations [31].

Enzyme kinetics set the forward capacity. The reference capacity is given by:

$$V^\circ_{\max,j} \;=\; k^\circ_{\mathrm{cat},j}\, e^\circ, \tag{13}$$

where $k^\circ_{\mathrm{cat},j}$ is the turnover number and e° is a reference enzyme concentration. BRENDA reports turnover numbers [32]. BioNumbers reports cellular abundance scales [33]. Together, these quantities set the reference capacity. The saturation factor gives the fraction of the reference capacity $V^\circ_{\max,j}$ available at the current substrate concentration. For a single-substrate enzyme, the reaction mechanism is written as:

$$E + S_j \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} [ES] \xrightarrow{k_\mathrm{cat}} E + P, \tag{14}$$

where E is free enzyme, S_j is substrate, $[ES]$ is the enzyme–substrate complex, and P is product. The rate constants k_1, k_{-1}, and k_cat describe substrate association, substrate dissociation, and catalysis, respectively. The quasi-steady-state approximation and enzyme conservation, $[E]_T = [E] + [ES]$, give $[ES] = [E]_T[S_j]/(K_{M,j} + [S_j])$. Here $[E]_T$ is total enzyme concentration, and $K_{M,j} = (k_{-1} + k_\mathrm{cat})/k_1$ is the Michaelis constant. With the reaction rate $v_j = k_\mathrm{cat}[ES]$ and maximum rate $V_{\max,j} = k_\mathrm{cat}[E]_T$, the Michaelis–Menten law is given by [5]:

$$v_j \;=\; V_{\max,j} \frac{[S_j]}{K_{M,j} + [S_j]} \qquad \Longrightarrow \qquad f_j([S_j]) \;=\; \frac{v_j}{V_{\max,j}} \;=\; \frac{[S_j]}{K_{M,j} + [S_j]} \;\in [0,1], \tag{15}$$

where $[S_j]$ is the substrate concentration. The saturation factor f_j approaches zero at low substrate and one at saturation. Because the saturation factor f_j depends on the substrate concentration $[S_j]$, the bound becomes state-dependent.

Gene expression sets enzyme abundance. The balances for transcript m_j and protein p_j are given by:

$$\dot{m}_j = r_{X,j}\, u_j - (\theta_{m,j} + \mu)\, m_j + \lambda_j, \tag{16}$$
$$\dot{p}_j = r_{L,j}\, w_j\, m_j - (\theta_{p,j} + \mu)\, p_j, \tag{17}$$

where $r_{X,j}$ and $r_{L,j}$ are transcription and translation capacities. The quantities u_j and w_j are the dimensionless transcriptional and translational controls, respectively. The quantities $\theta_{m,j}$ and $\theta_{p,j}$ are degradation constants. The term λ_j is the basal transcription rate. The growth rate μ adds to both loss rates. At steady state, the solutions are:

$$m_j^\star = \frac{r_{X,j} u_j + \lambda_j}{\theta_{m,j} + \mu}, \qquad p_j^\star = \frac{r_{L,j} w_j\, m_j^\star}{\theta_{p,j} + \mu}, \qquad (e/e^\circ) = \frac{p_j^\star}{e^\circ}. \tag{18}$$

Gene–protein–reaction rules map the steady-state protein abundance $p_j^\star$ to catalytic activity for single enzymes, multisubunit complexes, and isozymes. Measured transcript or protein abundance can instead set the enzyme-abundance ratio (e/e°) directly and produce condition-specific capacities.

Active and inactive configurations provide one statistical-mechanical description of allosteric, transcriptional, and translational control (Fig. 2). For an enzyme, promoter, or translation complex, the control is the statistical weight of the active set divided by the weight of all configurations. Let $\mathcal{S}_q$ be the configurations for process q, where q denotes catalysis, transcription, or translation. For a configuration $s \in \mathcal{S}_q$ with energy ϵ_s, the state weight, partition function, and state probability are given by:

$$\omega_s(\mathbf{x}) = g_s(\mathbf{x}) e^{-\beta\epsilon_s}, \qquad Z_q(\mathbf{x}) = \sum_{r\in\mathcal{S}_q} \omega_r(\mathbf{x}), \qquad p_s(\mathbf{x}) = \frac{\omega_s(\mathbf{x})}{Z_q(\mathbf{x})}, \tag{19}$$

where $\beta = 1/(k_B T_{\text{abs}})$ is the inverse thermal energy. The constant k_B is the Boltzmann constant. The quantity T_{abs} is the absolute temperature. The quantity Z_q is the partition function. The concentration factor $g_s(\mathbf{x})$ includes ligand concentration, binding stoichiometry, and degeneracy. For example, a state with binding stoichiometry n, effector concentration x, and concentration scale K has a concentration factor proportional to $(x/K)^n$.

Let $\mathcal{A}_q \subseteq \mathcal{S}_q$ contain the active configurations and let $\mathcal{I}_q = \mathcal{S}_q \setminus \mathcal{A}_q$ contain the inactive configurations. The control Φ_q is the probability of the active set:

$$\Phi_q(\mathbf{x}) = \sum_{s\in\mathcal{A}_q} p_s(\mathbf{x}) = \frac{\displaystyle\sum_{s\in\mathcal{A}_q} g_s(\mathbf{x}) e^{-\beta\epsilon_s}}{\displaystyle\sum_{r\in\mathcal{S}_q} g_r(\mathbf{x}) e^{-\beta\epsilon_r}} = \frac{Z_{\mathcal{A}_q}}{Z_{\mathcal{A}_q} + Z_{\mathcal{I}_q}}. \tag{20}$$

Here $Z_{\mathcal{C}} = \sum_{s\in\mathcal{C}} \omega_s$ is the partition sum over a configuration subset $\mathcal{C}$. Active states appear in the numerator and denominator. Inactive states appear only in the denominator. Therefore $0 \le \Phi_q \le 1$ for the control Φ_q. An effector increases or decreases the control Φ_q according to the states it stabilizes. Choose configuration 0 as a reference, define the energy difference $\Delta\epsilon_s = \epsilon_s - \epsilon_0$, and denote the effective relative weight of state s by $K_{\star,s}$. The relative weights are then given by:

$$K_{\star,s} \equiv \frac{\omega_s/g_s}{\omega_0/g_0} = e^{-\beta\Delta\epsilon_s}, \qquad \frac{\omega_s}{\omega_0} = \frac{g_s(\mathbf{x})}{g_0(\mathbf{x})} K_{\star,s}. \tag{21}$$

With the reference weight ω_0 and reference concentration factor g_0 set to one, the state weight is $\omega_s = g_s(\mathbf{x}) K_{\star,s}$. A term such as $K_2 f_{TL}$ combines the energetic factor $K_2 = e^{-\beta\Delta\epsilon_2}$ with the available ligand-bound transcription-factor fraction f_{TL}. Fitted state-weight parameters K_1 and K_2 may also include constant polymerase concentration, standard-state terms, or other fixed contributions. They need not be measured dissociation constants.

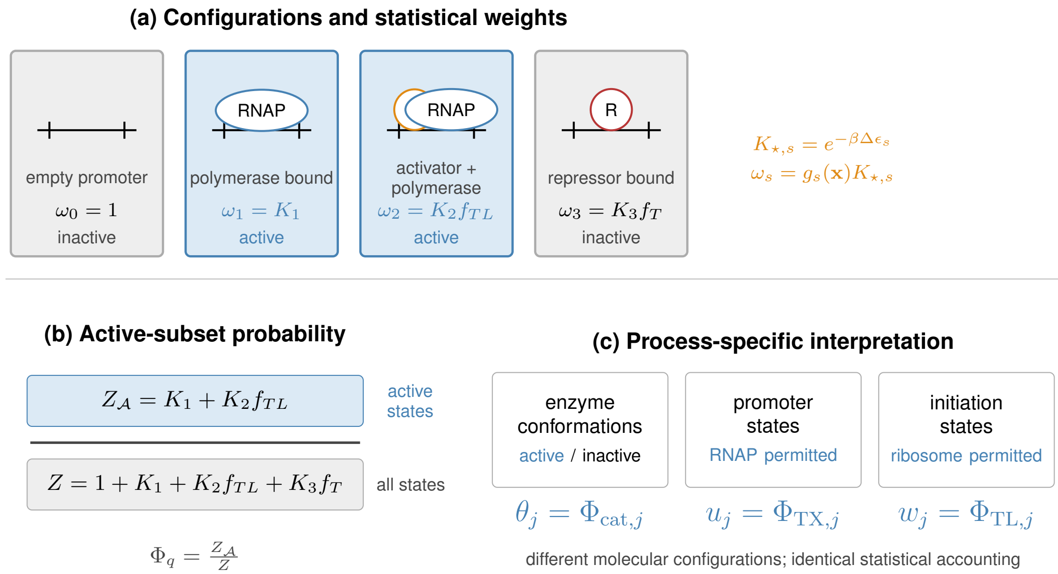

Figure 2: Active-subset construction of biological control factors. (a) Configuration weights combine relative energetics with regulator availability. Blue states permit the process and gray states do not. (b) Active states enter the numerator, while all states enter the partition-function denominator. The promoter shown is schematic. (c) Specifying the relevant configurations gives allosteric activity θ_j, transcriptional control u_j, or translational control w_j from the same statistical accounting.

Moon, Voigt, and coworkers used this construction for inducible promoters [34]. The Lux promoter has state weights 1, K_1, and $K_2 f_{TL}$ for the empty, polymerase-bound, and polymerase–LuxR-bound states. Both polymerase-bound states permit transcription. The Tac promoter has state weights 1, K_1, and $K_2 f_T$ for the empty, polymerase-bound, and repressor-bound states. Only the polymerase-bound state permits transcription. Normalization gives the promoter controls:

$$u_{\mathrm{Lux}} = \frac{K_1 + K_2 f_{TL}}{1 + K_1 + K_2 f_{TL}}, \qquad u_{\mathrm{Tac}} = \frac{K_1}{1 + K_1 + K_2 f_T}. \tag{22}$$

Here the fraction f_{TL} is ligand-bound transcription factor and the fraction f_T is ligand-free transcription factor. Increasing inducer adds weight to an active Lux state and removes weight from an inactive Tac state. Both controls are special cases of Equation (20). A two-state model gives a familiar special case. If the inactive state has weight $\omega_I = 1$ and the effector-stabilized active state has weight $\omega_A = (x/K)^n e^{-\beta\Delta\epsilon}$, then the control is given by:

$$\Phi_q(x) = \frac{(x/K)^n e^{-\beta\Delta\epsilon}}{1 + (x/K)^n e^{-\beta\Delta\epsilon}}, \tag{23}$$

where the energy difference $\Delta\epsilon$ is the active-state energy relative to the inactive state. If the effector instead stabilizes the inactive state, its weight appears only in the denominator and the control decreases with the effector concentration x.

The active-state control in Equation (20) maps directly to each process-specific factor:

$$\theta_j = \Phi_{\mathrm{cat},j}, \qquad \bar{u}_j = \Phi_{\mathrm{TX},j}, \qquad w_j = \Phi_{\mathrm{TL},j}. \tag{24}$$

The allosteric factor θ_j multiplies catalytic capacity in Equation (10). The total transcriptional control $\bar{u}_j$ controls transcription, and the translational control w_j controls translation in Equation (17). These controls are distinct from the degradation constants $\theta_{m,j}$ and $\theta_{p,j}$. Split the transcriptionally active set into regulated and basal states. Let $\mathcal{A}_{\mathrm{TX}}^{\mathrm{reg}}$ contain the regulated states and $\mathcal{A}_{\mathrm{TX}}^{\mathrm{basal}}$ contain states that support transcription without the regulated input. The resulting split is given by:

$$\bar{u}_j = \frac{Z_{\mathcal{A}_{\mathrm{TX}}^{\mathrm{reg}}}}{Z_{\mathrm{TX}}} + \frac{Z_{\mathcal{A}_{\mathrm{TX}}^{\mathrm{basal}}}}{Z_{\mathrm{TX}}} = u_j + u_j^{\dagger}, \qquad \lambda_j \equiv r_{X,j} u_j^{\dagger}. \tag{25}$$

Here Z_{TX} is the total transcriptional partition function. The regulated transcriptional control u_j enters Equation (16). The basal control $u_j^{\dagger}$ defines the basal transcription rate λ_j. Moon, Voigt, and coworkers used this promoter-state construction [34]. Adhikari and coworkers extended it to transcriptional and translational control [35].

These factors require thermodynamic, kinetic, expression, and regulatory data. When data are unavailable, set the saturation factor $f_j \approx 1$, the enzyme-abundance ratio $(e/e^{\circ}) \approx 1$, and the allosteric factor $\theta_j \approx 1$. This gives the baseline bound:

$$-\delta_j \, V_{\max,j}^{\circ} \;\leq\; \hat{v}_j \;\leq\; V_{\max,j}^{\circ}. \tag{26}$$

The baseline bound in Equation (26) retains only reversibility and reference capacity. The general bound in Equation (10) identifies the omitted factors that require additional data. Each factor has a precedent in constraint-based modeling: thermodynamics-based flux analysis uses free energies and metabolite activity ranges to exclude infeasible states [29]. Enzyme-constrained FBA uses turnover numbers and enzyme abundance. GECKO applies this approach at genome scale [36]. Metabolism-and-expression models couple transcription and translation to metabolism [37]. Regulatory FBA uses Boolean expression rules [38], whereas PROM uses data-derived interaction probabilities [39]. The allosteric factor θ_j represents direct regulation of enzyme activity. The general bound in Equation (10) places these established terms in one notation. The balances above use aggregate transcription and translation capacities $r_{X,j}$ and $r_{L,j}$. A sequence-resolved model can instead calculate nucleotide, amino-acid, and energy demands from each gene and protein sequence and add those reactions to the stoichiometric matrix $\mathbf{S}$ [11]. Expression then becomes part of the flux calculation. Vilkhovoy and coworkers used this construction for cell-free protein synthesis [40].

5 Worked Example: Urea-Cycle Metabolism

Thermodynamic and kinetic data set flux bounds for a urea-cycle network with four cycle reactions, a competing nitric oxide synthase reaction, and exchange with the environment (left panel, Fig. 3). The inputs come from public, cross-organism databases rather than HL-60-specific measurements. The model uses the steady-state mass-balance constraints $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$ from Equation (8) and the linear program in Equation (9). The application-specific quantities are the stoichiometric matrix $\mathbf{S}$, lower-bound vector $\boldsymbol{\ell}$, upper-bound vector $\mathbf{u}$, and objective coefficient vector $\mathbf{c}$. Argininosuccinate synthetase (v_1, EC 6.3.4.5) forms argininosuccinate. Argininosuccinate lyase (v_2, EC 4.3.2.1) produces arginine and fumarate. Arginase I (v_3, EC 3.5.3.1) produces ornithine and urea. Ornithine transcarbamylase (v_4, EC 2.1.3.3) regenerates citrulline. Nitric oxide synthase (v_5, EC 1.14.13.39) competes with arginase for arginine and produces nitric oxide and citrulline. The five enzymatic reactions involve eighteen metabolites. Fourteen metabolites cross the system boundary through exchange reactions b_1 through b_{14}. We use the secretion-positive convention $M_i \to \emptyset$ for boundary species M_i: positive flux denotes secretion, and negative flux denotes uptake. Arginine, citrulline, ornithine, and argininosuccinate remain inside the network. The resulting matrix has eighteen metabolite balances and nineteen reactions, so the stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{18\times 19}$.

The standard transformed reaction Gibbs energies from eQuilibrator and the heuristic direction assignments for the five enzymatic reactions are summarized under a common transformed-state convention (Table 1) [30, 31]. The estimates use pH 7.5, pMg 3.0, ionic strength 0.25 M, and absolute temperature $T_{\text{abs}} = 298.15$ K. The displayed uncertainties are eQuilibrator 95% confidence intervals. The reversibility threshold $\Delta_{\text{r}} G_j^{\prime *} = -10$ kJ/mol is a heuristic cutoff, not a value derived from a stated physiological activity range.

Table 1: Standard transformed reaction Gibbs energies from eQuilibrator at pH 7.5, pMg 3.0, ionic strength 0.25 M, and 298.15 K. The confidence intervals are the displayed 95% half-widths. Direction assignments use the heuristic threshold $\Delta_{\text{r}} G_j^{\prime *} = -10$ kJ/mol.

Reaction	EC number	$\Delta_{\text{r}} G_j^{\prime\circ}$ (kJ/mol)	95% CI (kJ/mol)	δ_j
v_1	6.3.4.5	−4.3	±2.9	1
v_2	4.3.2.1	11.6	±5.7	1
v_3	3.5.3.1	−33.9	±12.4	0
v_4	2.1.3.3	−30.3	±5.7	0
v_5	1.14.13.39	−1254.4	±29.6	0

The heuristic rule makes reactions v_1 and v_2 reversible and reactions v_3, v_4, and v_5 irreversible. Exchange reactions retain wide bounds in both directions because their direction depends on the medium. The kinetic capacities in Equation (13) combine nominal turnover numbers with the reference enzyme abundance $e^\circ = 0.01$ mmol/gDW. These parameters set an illustrative capacity scale rather than representing a matched set of HL-60 measurements. In the BRENDA database [32], queried from release 2026.1, the values for reactions v_2 through v_5 have exact turnover-number matches, although the selected records span organisms and assay conditions. Because the EC entry for reaction v_1 contains no turnover-number field, this reaction uses $k_{\text{cat}} \simeq 10\ \text{s}^{-1}$ as a default. This value is the median of 1,942 unique enzyme–natural-substrate reactions reported by Bar-Even et al. and is not an enzyme-specific measurement [41]. The complete record, including organisms, substrates, assay conditions, source publications, and selection rules, is provided in the accompanying data repository. The common reference abundance is also illustrative rather than a reaction-specific physiological measurement. BioNumbers provides cellular abundance scales that can guide its replacement when condition-specific measurements are available [33]. Argininosuccinate synthetase uses the default turnover number $k^\circ_{\text{cat}} = 10.0\ \text{s}^{-1}$, which gives the reference capacity $V^\circ_{\max,v_1} = 360\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$. Arginase I has the turnover number $k^\circ_{\text{cat}} = 190.0\ \text{s}^{-1}$, giving the reference capacity $V^\circ_{\max,v_3} = 6840\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$. Ornithine transcarbamylase has the turnover number $k^\circ_{\text{cat}} = 410.0\ \text{s}^{-1}$, giving the reference capacity $V^\circ_{\max,v_4} = 14760\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$. Argininosuccinate lyase has the lowest turnover number, $k^\circ_{\text{cat},v_2} = 3.28\ \text{s}^{-1}$, which gives the reference capacity $V^\circ_{\max,v_2} = 118.08\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$. This is the smallest of the first four capacities. Nitric oxide synthase has a BRENDA turnover number of $k^\circ_{\text{cat},v_5} = 1.08\ \text{s}^{-1}$ for L-arginine from rat NOS, giving $V^\circ_{\max,v_5} = 38.88\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$. This smaller branch capacity did not alter the nominal maximum-urea solution, in which v_5 carried zero flux. Exchange reactions retain wide capacity bounds.

The model has no substrate, expression, or regulatory data. We therefore set the saturation factor f_j, enzyme-abundance ratio (e/e°), and allosteric factor θ_j to one. These settings yield the reported optimal flux distribution (right panel, Fig. 3). The enzymatic bounds therefore reduce to $-\delta_j V^\circ_{\max,j} \le \hat{v}_j \le V^\circ_{\max,j}$ for reactions v_1 through v_5. The fourteen exchange reactions use wide bounds of $-1000 \le \hat{v}_j \le 1000$ mmol gDW^{-1} h^{-1}. The objective is urea export. Because exchange reaction b_4 is written as urea $\to \emptyset$, we set its objective coefficient $c_{b_4} = 1$ and all other entries of the objective vector $\mathbf{c}$ to zero. The linear program [9] gave a unique optimum. Reactions v_1 through v_4 each carried 118.08 mmol gDW^{-1} h^{-1}, the capacity of argininosuccinate lyase. The competing nitric oxide synthase branch v_5 carried zero flux because the objective assigns all available arginine to urea production. The optimal urea export was 118.08 mmol gDW^{-1} h^{-1} at the urea-export flux $\hat{v}_{b_4} = 118.08$. Every exchange flux carrying carbon or nitrogen into or out of the cycle scaled to match it. In particular, the uptake of carbamoyl phosphate and aspartate and the secretion of urea and fumarate each had magnitude 118.08 mmol gDW^{-1} h^{-1}. Flux-variability analysis gave the same minimum and maximum for every reaction when the objective was held at its optimum. The optimal flux distribution is therefore unique to numerical tolerance.

The Monte Carlo analysis treats the database inputs as uncertain rather than exact. It also compares the effect of imputing the missing saturation factor (Fig. 4). The calculation uses a separate illustrative free-energy spread rather than the eQuilibrator confidence intervals in Table 1. We assigned lognormal uncertainty of about a factor of two to each turnover number and the reference abundance and independent normal uncertainty with standard deviation $\sigma_{\Delta G} = 2.0$ kJ/mol to each standard transformed reaction Gibbs energy. For four reactions, we also sampled substrate concentrations and Michaelis constants from Park and coworkers [42]. Argininosuccinate lyase lacked a measured substrate concentration, so its saturation factor remained at one. We solved the linear program for ten thousand parameter draws using Algorithm 1. This Monte Carlo calculation produced a right-skewed urea-export distribution with median 105 and mean 156 mmol gDW^{-1} h^{-1} and a central ninety-five percent interval of 17 to 614. Varying only capacity and Gibbs-energy inputs gave a median of 113 mmol gDW^{-1} h^{-1}. Adding the four measured saturation factors changed the median to 105, so those factors had little effect. Most of the uncertainty came from the sampled capacity of argininosuccinate lyase. The reversibility switches did not bind because the optimal cycle flux was forward and the other four enzymes usually retained excess capacity. The missing saturation factor for argininosuccinate lyase had a larger effect: imputing it from a physiological argininosuccinate concentration and a BRENDA Michaelis constant lowered the median to 50 mmol gDW^{-1} h^{-1} and widened the distribution toward zero. Better data for this limiting reaction would therefore reduce the uncertainty most directly. Because no HL-60-specific concentration, expression, or regulatory measurements were available, eQuilibrator provided the standard transformed reaction Gibbs-energy estimates. The heuristic threshold set reaction direction. BRENDA set the capacity scales. The other factors remained at one. Under these assumptions, argininosuccinate lyase reaction v_2 limited the result. The competing nitric oxide synthase reaction v_5 carried little flux.

6 Worked Example: A Feedback-Inhibited Linear Pathway

The urea-cycle bounds used thermodynamic and kinetic factors but set the enzyme-abundance ratio (e/e°) and allosteric factor θ_j to one. The feedback-inhibited pathway includes both controls. Its end product inhibits the existing enzyme and represses transcription of its gene, thereby changing enzyme activity and enzyme abundance on different timescales. These coupled effects are

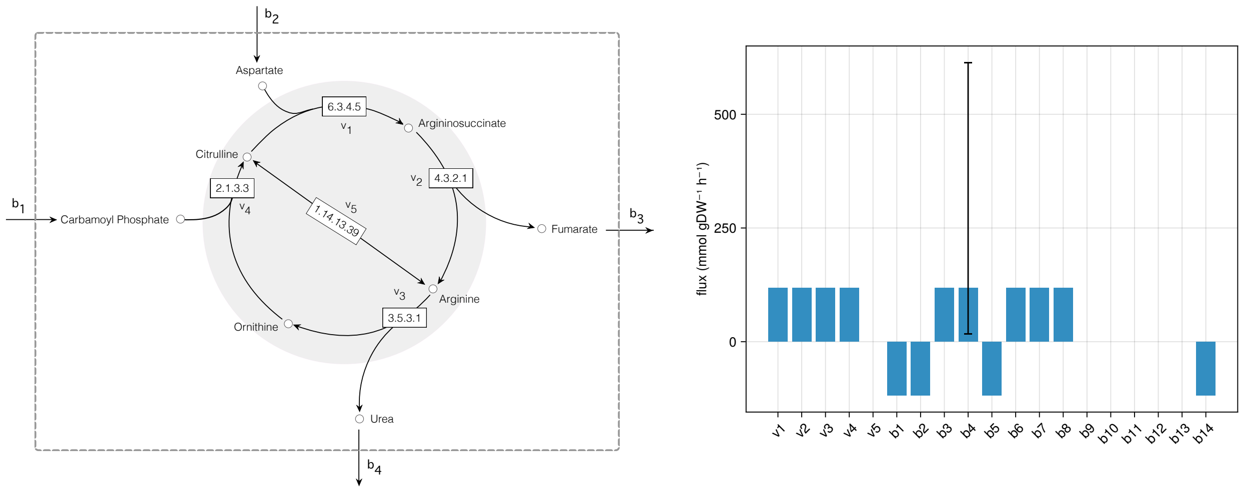

Figure 3: Urea-cycle network and optimal fluxes. Left: reactions v_1 through v_4 form the cycle. Reaction v_5 is the competing nitric oxide synthase branch. Exchange reactions cross the dashed boundary. Arginine, citrulline, ornithine, and argininosuccinate remain internal. Right: the optimal flux distribution for Equation (9). Positive exchange flux denotes secretion, and negative exchange flux denotes uptake. The cycle reactions carried 118.08 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$, the capacity of argininosuccinate lyase (v_2). Nitric oxide synthase reaction v_5 carried zero flux. The whisker on urea export b_4 shows the 2.5–97.5 percentile interval from Configuration A of the Monte Carlo calculation in Algorithm 1.

Algorithm 1 Monte Carlo propagation of urea-cycle parameter uncertainty

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Require: nominal turnover numbers k°_{cat}, reference enzyme abundance e°, and standard transformed reaction Gibbs energies $\Delta_{\mathrm{r}}G'^{\circ}$; nominal substrate concentrations $\overline{[S_j]}$ and Michaelis constants $\overline{K_{M,j}}$ where measured; log-space spread $\sigma_{\ln}$ and free-energy standard deviation $\sigma_{\Delta G}$; reversibility threshold $\Delta_{\mathrm{r}}G'^{*}$; draw count N; random seed

seed the random generator
for draw $i = 1$ to N **do**
  sample enzyme abundance $e \leftarrow e^{\circ}\exp(\sigma_{\ln} Z_e)$ ▷ one shared abundance draw, $Z_e \sim \mathcal{N}(0,1)$
  for each enzymatic reaction j **do**
    $k_{\mathrm{cat},j} \leftarrow k^{\circ}_{\mathrm{cat},j}\exp(\sigma_{\ln} Z_{k,j})$, $\quad \Delta_{\mathrm{r}}G'_j \leftarrow \Delta_{\mathrm{r}}G'^{\circ}_j + \sigma_{\Delta G} Z_{\Delta G,j}$ ▷ independent $Z_{k,j}, Z_{\Delta G,j} \sim \mathcal{N}(0,1)$ per reaction
    set reversibility factor $\delta_j \leftarrow \not\mathbb{1}[\Delta_{\mathrm{r}}G'_j > \Delta_{\mathrm{r}}G'^{*}]$
    if substrate data measured for reaction j **then**
      $[S_j] \leftarrow \overline{[S_j]}\exp(\sigma_{\ln} Z_{S,j})$, $\quad K_{M,j} \leftarrow \overline{K_{M,j}}\exp(\sigma_{\ln} Z_{K,j})$ ▷ independent $Z_{S,j}, Z_{K,j} \sim \mathcal{N}(0,1)$
      set saturation factor $f_j \leftarrow [S_j]/(K_{M,j} + [S_j])$
    else
      set saturation factor $f_j \leftarrow 1$
    end if
    set capacity $V_{\max,j} \leftarrow 3600\,k_{\mathrm{cat},j}\,e$; upper bound $b^{\mathrm{U}}_j \leftarrow V_{\max,j} f_j$; lower bound $b^{\mathrm{L}}_j \leftarrow -\delta_j V_{\max,j} f_j$
  end for
  $\hat{\mathbf{v}}^{(i)} \leftarrow \arg\max\{\mathbf{c}^{\top}\mathbf{v} : \mathbf{S}\mathbf{v} = \mathbf{0},\ \mathbf{b}^{\mathrm{L}} \le \mathbf{v} \le \mathbf{b}^{\mathrm{U}}\}$
end for
return per-reaction mean, standard deviation, and 2.5/50/97.5 percentiles of $\{\hat{\mathbf{v}}^{(i)}\}_{i=1}^{N}$

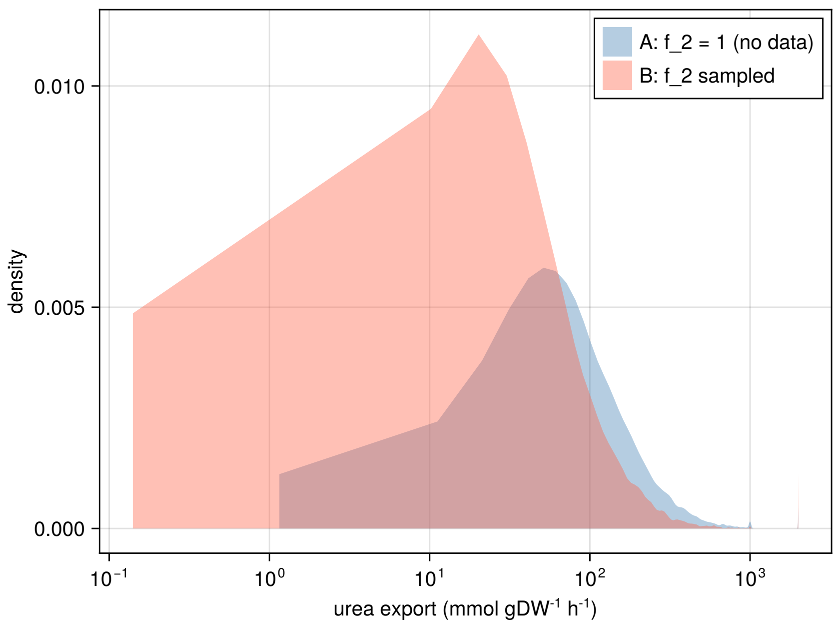

Figure 4: Sensitivity of urea export to saturation information. Configuration A (blue) sampled four saturation factors derived from data reported by Park and coworkers [42]. It held the saturation factor $f_2 = 1$ at the argininosuccinate lyase bottleneck because its substrate is unmeasured. The resulting median urea export was 105 $\text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$. Configuration B (red) imputed the saturation factor f_2 from a physiological argininosuccinate concentration and a BRENDA Michaelis constant and sampled it as well. This shifted the median to 50 $\text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$ and widened the distribution toward zero. The horizontal axis is logarithmic.

represented in a linear pathway with one controlled step:

$$\emptyset \xrightarrow{r_0} X_1 \xrightarrow{r_1} X_2 \xrightarrow{r_2} X_3 \xrightarrow{r_3} \emptyset, \tag{27}$$

Here r_0 is the committed uptake step. Reactions r_1 and r_2 are interconversions. Reaction r_3 exports the end-product metabolite X_3. This end product controls both the activity and abundance of the enzyme for committed step r_0. We generated a kinetic reference with a Biochemical Systems Theory (BST) S-system [6, 7]. The model includes five dynamic states: metabolites X_1, X_2, and X_3, transcript m, and enzyme E_0. BSTModelKit.jl was used to integrate these states to steady state [8]. The end product enters the committed-step rate with kinetic order $-a$ and the transcription rate with kinetic order $-b$. Increasing the end-product concentration X_3 therefore lowers both enzyme activity and enzyme production. Metabolic fluxes obey the steady-state balance $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$, while transcript and protein balances include degradation and growth dilution. The committed-step rate v_{r_0} and BST activity factor θ_{BST} are given by:

$$\begin{aligned} v_{r_0} &= V^{\circ}_{\max,r_0}\,(e/e^{\circ})\,\theta_{\mathrm{BST}}, \\ \theta_{\mathrm{BST}} &= (X_3/X_3^{\circ})^{-a}, \qquad a = 0.4, \\ \text{transcription} &\propto (X_3/X_3^{\circ})^{-b}, \qquad b = 0.6, \end{aligned} \tag{28}$$

where the reference end-product concentration $X_3^{\circ} = 1$ in the concentration units used by the simulation. The normalized ratios are dimensionless.

The kinetic reference combined expression timescales with metabolic feedback and gave a steady-state pathway throughput $T^{\star} \approx 2.66$ (Fig. 5). The simulation used a 330-residue enzyme, a 990-nucleotide gene, polymerase elongation at 60 nucleotides per second, and ribosome elongation at 16 amino acids per second. These values provided illustrative expression timescales. The steady-state abundance also depended on an mRNA half-life of 2.5 minutes, a protein half-life of 35 minutes, and a culture doubling time of 60 minutes. The same growth rate μ appears in both balances, but its relative contribution differs because transcript and protein have different degradation rates. With transcript degradation rate θ_m and protein degradation rate θ_p, growth accounted for about four percent of the total transcript clearance rate $\theta_m + \mu \approx 0.289\ \mathrm{min}^{-1}$ and about thirty-seven percent of the total enzyme clearance rate $\theta_p + \mu \approx 0.031\ \mathrm{min}^{-1}$. At steady state, every reaction in the linear chain carries the same pathway throughput T. Because export is first order in the end-product concentration X_3, the concentration is $X_3 = T/k_3$, where k_3 is the first-order export constant. Substitution of the activity and expression controls gives one consistency condition for the end-product concentration X_3. The numerical solution gave the steady-state concentration $X_3^{\star} \approx 4$. At this solution, the activity factor was $\theta^{\star}_{\mathrm{BST}} \approx 0.574$, and the enzyme-abundance ratio was $(e/e^{\circ})^{\star} \approx 0.464$. The basal transcription rate λ, defined by Equations (16) and (25), kept expression above zero. The kinetic orders $a = 0.4$ and $b = 0.6$ made both controls affect the result. Neither control alone produced the reference throughput.

The FBA comparison keeps the chain's stoichiometry unchanged and places the entire dual-control content on the committed-step bound (Fig. 5). The bound on committed-step flux $\hat{v}_{r_0}$ is given by:

$$0 \;\leq\; \hat{v}_{r_0} \;\leq\; V^{\circ}_{\max,r_0}\,(e/e^{\circ})\,\theta_{\mathrm{FBA}}(X_3), \qquad V^{\circ}_{\max,r_0} = 10, \tag{29}$$

which is the expression-and-regulation form of Equation (10). The thermodynamic and saturation factors remain at one. To avoid copying the BST rate law into the FBA bound, we represent the FBA activity factor θ_{FBA} with a bounded two-state Hill function using half-saturation scale $K = 5$ and Hill coefficient $n = 2$:

$$\theta_{\mathrm{FBA}}(X_3) \;=\; \frac{K^n}{K^n + X_3^n}, \qquad K = 5, \quad n = 2, \tag{30}$$

the repressive form of Equation (23). The active state has unit weight, and the inactive state has weight $(X_3/K)^n$. The half-saturation scale K and Hill coefficient n were specified independently of the BST kinetic order a. Both control factors were evaluated from the steady-state enzyme abundance and end-product concentration X_3, not from the reference flux. Four FBA solutions used the same stoichiometric matrix and differed only in the bound on committed step r_0. With both the expression and activity factors set to one, the predicted flux was 10. Expression control alone reduced it to 4.64. Activity control alone reduced it to 6.10. Both controls reduced it to 2.83. The dual-control result was within about six percent of the BST reference value, 2.66.

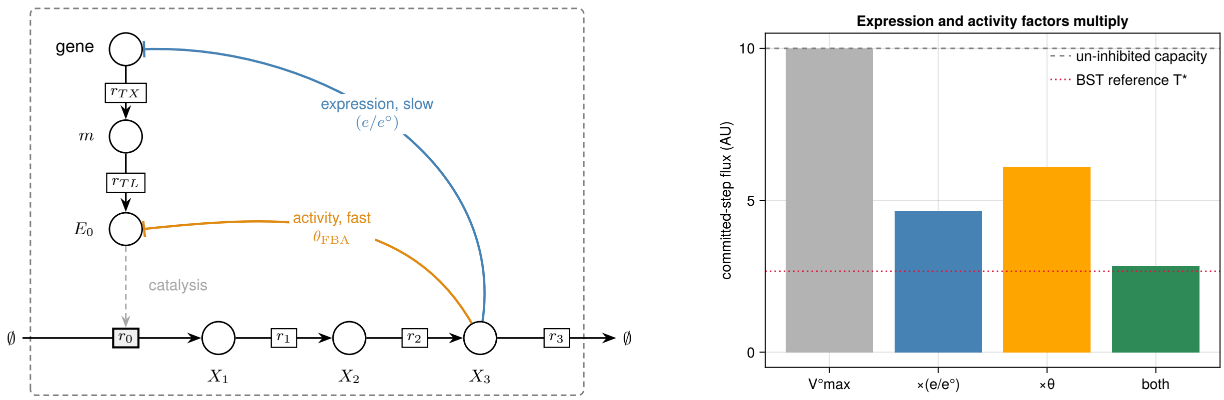

Figure 5: Dual-control feedback example. Left: the pathway and the transcription and translation steps that produce enzyme E_0. The end product X_3 represses enzyme expression and inhibits enzyme activity. Right: predicted committed-step flux r_0 with neither control, each control alone, and both controls. Expression control gave 4.64. Activity control gave 6.10. Both controls gave 2.83. The BST reference is pathway throughput $T^\star \approx 2.66$. The dashed line marks the uninhibited capacity of 10. All fluxes are in arbitrary units.

The parameter sweep addresses uncertainty in the Hill parameters, which were specified as half-saturation scale $K = 5$ and Hill coefficient $n = 2$ rather than fitted (Fig. 6). We therefore varied the scale K from 3 to 8 and coefficient n from 1 to 3. We recomputed the FBA activity factor θ_{FBA} at the reference end-product concentration X_3 and solved the program at each grid point. With both controls, the predicted flux ranged from about 1.4 to 4.1 and included the BST value of 2.66. The nominal parameters gave 2.83. Activity control alone kept the flux above about 2.97. Expression control alone kept it at 4.64. Neither single-control model included the BST value over the tested parameter range. The dual-control model did.

7 Integration in Practice

Small networks make the effects of individual bounds transparent, whereas practical applications combine many more reactions and data types. Dynamic cell-free protein synthesis provides one such application by coupling metabolism, expression, and regulation. Vilkhovoy and coworkers built a sequence-specific model of *E. coli* cell-free protein synthesis [40]. Transcription and translation reactions derived from DNA and mRNA sequences appear in the stoichiometric matrix with metabolism. Promoter and ribosome control functions u_j and w_j follow the biophysical models of Moon, Voigt, Adhikari, and coworkers [34, 35]. Dai, Horvath, and Varner extended the sequence-specific formulation to a dynamic constraint-based model that solved an optimization problem over

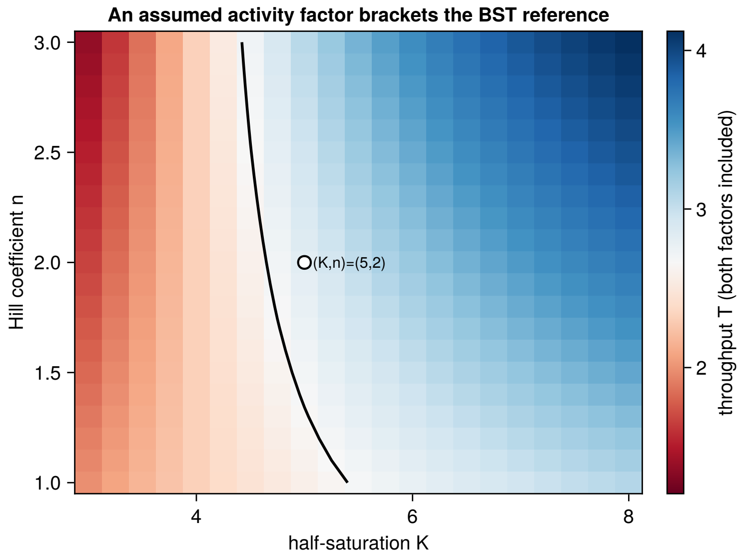

Figure 6: Sensitivity to the assumed activity-factor shape. The plot shows the dual-control flux while the half-saturation scale K and Hill coefficient n vary in the FBA activity factor $\theta_{\text{FBA}} = K^n/(K^n + X_3^n)$. The color scale is centered on the BST reference $T^\star \approx 2.66$. The black contour marks equal FBA and BST fluxes. The predicted flux ranged from about 1.4 to 4.1. The marked nominal parameter pair, $(K, n) = (5, 2)$, gave 2.83. All fluxes are in arbitrary units.

short time intervals while enforcing time-dependent metabolite constraints [25]. A later integrated model updated thermodynamic, kinetic, expression, and regulatory bounds during the reaction and was compared with time courses for 63 metabolites, mRNA, protein, and enzyme activity [26]. Cell-free batches have no growth dilution, so the growth rate $\mu = 0$. Their molecular pools still change. On a fixed concentration basis, their balance is therefore the dynamic form $\mathbf{Sv} = d\mathbf{x}/dt$ for cell-free flux vector $\mathbf{v}$, not the steady-state form $\mathbf{Sv} = \mathbf{0}$.

Wayman and coworkers used a constraint-based model for strain design. They added the N-linked glycosylation pathway of *Campylobacter jejuni* onto a genome-scale reconstruction of *E. coli* metabolism and searched for gene deletions that couple designer-glycan production to growth [27]. Each candidate knockout sets a reaction bound to zero, after which the linear program is solved again. Enzyme attenuation or overexpression can instead decrease or increase the enzyme-abundance ratio (e/e°) in Equation (10). The single Δgnd deletion, drawn from the model-identified strain families, increased measured glycan production nearly threefold relative to wild type. The model identified candidate perturbations. It did not predict the magnitude of this experimental increase. The two applications use bounds for different purposes: the cell-free model uses measurements to estimate fluxes, whereas the glycan model changes bounds to test possible interventions. Strain design also requires a discrete search over deletions or expression changes, which can be formulated as a bilevel program [28].

8 Outlook

Standard transformed reaction Gibbs energies, turnover numbers, and regulatory models set the factors in Equation (10). Time-course data can instead be used to estimate turnover numbers, saturation constants, allosteric factors θ_j, and expression controls u_j and w_j. Tabulated values remain useful as prior information when dynamic data are unavailable. The same approach can be applied reaction by reaction in genome-scale models. These models have many flux distributions that satisfy the steady-state balance $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$, so condition-specific bounds are especially important. Setting an unknown factor to one preserves feasibility but may leave many fluxes unresolved. Time-course data also allow models to retain accumulation and growth dilution. Cell-free systems have no growth dilution, but their molecular pools can still accumulate. Growing cultures can exhibit both accumulation and growth dilution. When measurements resolve these dynamics, the dynamic balance in Equation (6) provides the appropriate mass-balance constraints. For a fixed-basis cell-free batch, it reduces to the accumulation balance $\mathbf{S}\mathbf{v} = d\mathbf{x}/dt$ for cell-free flux vector $\mathbf{v}$ and concentration vector $\mathbf{x}$. We used this form in our earlier dynamic constraint-based model [25]. The steady-state approximation should be used only when accumulation and growth dilution are negligible. Before condition-specific bounds are used for design, their predictions should be tested against measurements not used to construct them, such as exchange rates, ^{13}C-derived fluxes, or responses to genetic and environmental perturbations.

Code and data availability

Source code, generated data, and instructions for reproducing the worked examples and figures are available at `https://github.com/varnerlab/MRW-BTC4-Chapter-Varner`. The repository includes the Julia project and manifest files needed to reproduce the computational environment.

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